Research Article

Coach WAR: A Counterfactual Approach to Evaluating NFL Head Coach Impact in the Modern Era

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Abstract: Evaluating NFL head coach performance is challenging due to complex interactions among coaching decisions, player talent, and organizational factors. We introduce a counterfactual framework measuring head coach impact by predicting team performance under a replacement-level coach, then comparing this baseline to actual results. This difference represents coaching Wins Above Replacement (WAR). Analyzing 1,637 team-seasons (1970–2024) with 365 features spanning draft history, injuries, rosters, team performance, and coaching histories, we employ XGBoost regression to isolate coaching contributions from team context. The model achieves test $R^2 = 0.602$ on held-out coaches, with a residual standard error of ± 2.1 wins per season. Results indicate meaningful variation in coaching impact: top-ranked coaches average up to +2.3 WAR per season (95% CI: [+1.3, +3.2]) while bottom-ranked coaches average −3.3 WAR (95% CI: [−4.7, −2.0]). Offensive versus defensive coaching backgrounds show a historical shift: defensive coaches held a significant advantage in the 1970s (−0.87, 95% CI: [−1.45, −0.29], $p = 0.005$), while offensive coaches average +0.59 more in the 2020s (95% CI: [−0.05, +1.24], $p = 0.11$). Year-to-year persistence analysis reveals low stability, with WAR explaining only 6.9% of next-season variance. A quintile analysis shows that top-quintile coaches lose approximately 79% of their advantage by the following season, consistent with scheme obsolescence as opponents adapt. Offensive coaches show significantly less multi-year persistence than defensive coaches ($p = 0.009$).

Keywords: coaching analytics, wins above replacement, NFL, sports analytics

1 Introduction

Evaluating head coach performance in the National Football League (NFL) presents unique challenges due to complex interactions among coaching decisions, player talent, and organizational factors. Traditional metrics such as win–loss records conflate team success with coaching quality, failing to isolate a coach's contribution from the context in which that coach operates.

We propose a counterfactual framework for measuring head coach impact: for each team-season, we predict how the team would have performed under a replacement-level coach with league-median characteristics, then compare this prediction to actual results. The difference represents coaching Wins Above Replacement (WAR). By controlling for roster quality, strength of schedule, salary allocation, injury exposure, and other contextual factors, the framework isolates the coaching contribution from the surrounding talent and organizational context.

2 Literature Review

Prior research on coaching evaluation has taken several approaches. Hadley et al. (2000) used a Poisson regression framework with basic team statistics to assess NFL coaching efficiency relative to production frontiers, while Goff and Wisley (2006) examined managerial lifecycle effects. The present work expands on

Hadley et al. (2000) by isolating coaching impact from a broader set of contextual factors, including salary distribution, player turnover, injury data, and detailed coach-specific performance histories.

WAR methodologies for player evaluation are well established in baseball (Palmer and Thorn, 1984; Smith, 2008; Woolner, 2002) and have been extended with formal uncertainty quantification. Baumer et al. (2015) developed openWAR with resampling-based interval estimates, advocating for uncertainty quantification as standard practice in WAR metrics. Brill and Wyner (2024) proposed Grid WAR as an alternative aggregation for pitchers, further demonstrating the importance of methodological choices in WAR construction. In football, Yurko et al. (2019) introduced nflWAR for offensive player evaluation, and Sabin (2021) estimated player value using plus–minus models that isolate individual contributions within a team sport.

Tree-based machine learning methods have proven effective for NFL prediction tasks. Lock and Nettleton (2014) used random forests to estimate within-game win probability, demonstrating the effectiveness of ensemble tree methods for NFL prediction tasks. More broadly, the growth of NFL tracking data has expanded the scope of quantitative football research (Lopez, 2020). In basketball, Deshpande and Jensen (2016) estimated individual player impact on team winning probability, addressing challenges of attribution in team sports that parallel those in coaching evaluation.

Our contribution extends the WAR framework to coaching evaluation, adapting the counterfactual approach to isolate coaching impact in a setting where coaches influence outcomes indirectly through strategic and personnel decisions rather than direct play execution.

3 Methods

3.1 Data Sources and Feature Engineering

We analyzed 1,637 team-seasons spanning 1970–2024, representing all 32 current NFL franchises across the Modern Era of professional football. The analysis incorporated 365 features from multiple data sources: draft pick history, injury reports, starter designations, complete rosters, historical team performance, and comprehensive head coach career histories from Pro Football Reference, along with salary cap allocation and contract details from Spotrac. Feature counts by category are shown in Appendix A (Table 6).

The coach-specific feature set comprised 139 features in three groups: (1) seven coaching experience variables (years of NFL and college experience as position coach, coordinator, or head coach, plus number of prior head coaching hires); (2a) 33 normalized team performance statistics from the coach's most recent offensive coordinator tenure, and (2b) 33 from the most recent defensive coordinator tenure. Because no coach in our sample served as both offensive and defensive coordinator prior to becoming a head coach, exactly one of these two groups is always absent for each coach; and (3) 66 normalized statistics from the coach's head coaching history, comprising 33 team offensive metrics and 33 opposing-team offensive metrics (i.e., defensive performance). All performance features were z-score normalized within each season to ensure cross-era comparability. Non-coaching features captured roster construction (turnover rates, player age and experience by position), financial management (salary cap allocation percentages), injury exposure (games missed by starters), draft capital investment, and team performance context (strength of schedule, penalty rates). Data availability varied by category: team performance and roster metrics extended across all seasons, while salary cap data was limited to 2011–2024 and player injury data to 2009–2024.

3.2 The Counterfactual Framework

To isolate coaching impact from roster quality, organizational context, and other factors, we developed a counterfactual approach: for each team-season, we predict what the team's performance would have been under a replacement-level coach with league-median coaching characteristics, then compare this prediction to actual results.

Formally, let $\mathbf{x}_i = (\mathbf{x}_i^{(c)}, \mathbf{x}_i^{(t)})$ denote the feature vector for team-season i, where $\mathbf{x}_i^{(c)} \in \mathbb{R}^{139}$ are coaching features and $\mathbf{x}_i^{(t)} \in \mathbb{R}^{226}$ are non-coaching features. We compute replacement-level coaching features $\tilde{\mathbf{x}}^{(c)}$ as follows: for each coach j, we first compute the career mean $\bar{\mathbf{x}}_j^{(c)}$ across all of coach j's seasons, then take the component-wise median across all coaches:

$$\tilde{x}_k^{(c)} = \text{median}_j\big(\bar{x}_{j,k}^{(c)}\big), \quad k \in \{1, \ldots, 139\}. \tag{1}$$

This two-stage aggregation ensures each coach is weighted equally regardless of tenure length. The replacement-level prediction is then $\hat{y}_i^{(r)} = f(\tilde{\mathbf{x}}^{(c)}, \mathbf{x}_i^{(t)})$, where f is the trained model.

This approach parallels established WAR methodologies in baseball (Baumer et al., 2015; Palmer and Thorn, 1984; Woolner, 2002), football (Yurko et al., 2019), and basketball (Berri et al., 2006; Deshpande and Jensen, 2016), where player contributions are measured against replacement-level alternatives. Unlike baseball's replacement level, set at a .294 winning percentage representing readily available minor league talent (Appelman and Cameron, 2013; Baseball-Reference.com, n.d.), we define replacement-level coaching as the league median. This choice reflects the coaching labor market's structure: NFL head coaches average only 3.2 years of tenure (Goff and Wisley, 2006), substantially shorter than MLB players' average career of 5.6 years (Witnauer et al., 2007), making the median coach a more appropriate baseline than a theoretical minimum-quality hire.

3.3 Model Specification

We employed XGBoost (Chen and Guestrin, 2016), a gradient-boosted regression tree ensemble that minimizes the regularized objective

$$\mathcal{L} = \sum_{i=1}^{n} \ell\big(y_i, \hat{y}_i\big) + \sum_{m=1}^{M} \Omega(f_m), \quad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda\|w\|^2 + \alpha\|w\|_1, \tag{2}$$

where ℓ is the squared-error loss, T is the number of leaves in tree f, w are leaf weights, and γ, λ, α are regularization parameters. XGBoost was selected for its ability to capture non-linear relationships and feature interactions while providing interpretable feature importance metrics, properties also leveraged in NFL analytics by Lock and Nettleton (2014) for win probability estimation.

Missing values arising from incomplete early-era salary and injury data were imputed via truncated Singular Value Decomposition (SVD) matrix completion (Brand, 2002). The decomposition iterated until convergence (tolerance 10^{-4}), and non-normalized features were standardized prior to imputation to prevent scale-dependent bias.

To prevent data leakage, we employed coach-based train/test stratification: 216 coaches and all their associated seasons were assigned to the training set (81.7% of observations), while 54 held-out coaches formed the test set (18.3%). This ensures the model's test performance reflects generalization to entirely unseen coaching tenures. Hyperparameters were selected via randomized search over a discrete grid with 200 iterations and 5-fold cross-validation on the training set, optimizing R^2. Final hyperparameter values are shown in Appendix B (Table 7).

3.4 WAR Calculation

Coach WAR is computed as the difference between actual winning percentage and the replacement-level prediction, scaled to a 16-game season:

$$\text{WAR}_i = 16 \times \big(y_i - \hat{y}_i^{(r)}\big), \tag{3}$$

where y_i is the actual winning percentage and $\hat{y}_i^{(r)} = f(\tilde{\mathbf{x}}^{(c)}, \mathbf{x}_i^{(t)})$ is the replacement-level prediction from Equation 1. All WAR calculations are standardized to a 16-game season, including the 17-game seasons beginning in 2021, to ensure cross-era comparability and prevent artificial inflation of recent values. For 17-game seasons, this represents a 6% conservative adjustment.

3.5 Uncertainty Quantification

As advocated by Baumer et al. (2015) for player WAR, we report interval estimates alongside point estimates. Because coaching WAR is observed at the season level without sub-seasonal granularity, we use analytical t-intervals rather than the bootstrap resampling approach used for play-level player metrics. All p-values are two-sided unless otherwise noted.

3.5.1 Single-Season Precision

We report the model residual standard error, computed as the root mean squared error on the held-out coach test set converted to a 16-game season scale:

$$\mathrm{SE}_{\mathrm{model}} = 16 \times \mathrm{RMSE}_{\mathrm{test}}. \tag{4}$$

This provides a global measure of single-season WAR precision applicable to all observations. Because XGBoost does not produce observation-level prediction intervals, we treat $\mathrm{SE}_{\mathrm{model}}$ as a uniform bound on single-season estimation uncertainty.

3.5.2 Career WAR Intervals

For coach c with n_c seasons of individual WAR values $\{w_{c,1}, \ldots, w_{c,n_c}\}$, the career average WAR and its 95% confidence interval are

$$\overline{w}_c \pm t_{n_c-1,\,0.025} \cdot \frac{s_c}{\sqrt{n_c}}, \tag{5}$$

where s_c is the sample standard deviation of individual-season WAR values and $t_{n_c-1,\,0.025}$ is the critical value from the t-distribution with $n_c - 1$ degrees of freedom. These intervals directly address career sample size: coaches with few seasons (e.g., $n_c = 3$) produce wide intervals, while long-tenured coaches yield narrow intervals.

3.5.3 Decade Comparisons

To assess whether offensive or defensive coaching backgrounds produce systematically different WAR outcomes, we compare group means within each decade using two complementary tests. First, Welch's t-test (unequal variances) yields a t-statistic and p-value for the null hypothesis that the two group means are equal. The 95% confidence interval on the difference in means is

$$(\bar{w}_O - \bar{w}_D) \pm t_{\nu,\,0.025} \cdot \sqrt{\frac{s_O^2}{n_O} + \frac{s_D^2}{n_D}}, \tag{6}$$

where subscripts O and D denote offensive and defensive background groups, and ν is the Welch–Satterthwaite approximation to degrees of freedom:

$$\nu = \frac{\left(\frac{s_O^2}{n_O} + \frac{s_D^2}{n_D}\right)^2}{\frac{(s_O^2/n_O)^2}{n_O-1} + \frac{(s_D^2/n_D)^2}{n_D-1}}. \tag{7}$$

Second, the Mann–Whitney U test provides non-parametric confirmation by testing whether one group's WAR values tend to rank higher than the other's, without distributional assumptions. Effect size is reported as Cohen's d using the pooled standard deviation.

To quantify whether the offensive-versus-defensive gap has shifted over time, we fit a separate ordinary least squares regression of the per-decade mean difference ($\bar{w}_O - \bar{w}_D$) on decade midpoint. The slope estimates the change in the gap per decade (in games per season), and its p-value tests whether the trend is significantly different from zero.

3.5.4 Persistence Analysis

To evaluate year-to-year stability of coaching performance, we compute Pearson and Spearman correlation coefficients between consecutive-season WAR values within each coaching background group. To test whether persistence differs between offensive and defensive coaches, we apply Fisher's Z-transformation:

$$Z = \frac{\frac{1}{2}\ln\left(\frac{1+r_O}{1-r_O}\right) - \frac{1}{2}\ln\left(\frac{1+r_D}{1-r_D}\right)}{\sqrt{\frac{1}{n_O-3} + \frac{1}{n_D-3}}}, \tag{8}$$

where r_O and r_D are the Pearson correlations for the offensive and defensive groups, n_O and n_D are the respective sample sizes, and Z is compared to a standard normal distribution.

4 Results

The final model achieved $R^2 = 0.789$ on the training set and $R^2 = 0.602$ on held-out coaches (54 of 270 coaches withheld with all their seasons). The train–test gap of 0.19 indicates moderate overfitting, mitigated by regularization (max depth = 2, ℓ_1/ℓ_2 penalties). The model residual standard error on the test set corresponds to ±2.1 wins per 16-game season (Section 3.5), providing a baseline for interpreting single-season WAR precision. To verify findings are not artifacts of the XGBoost algorithm, we compared coach WAR estimates to those from Ridge regression trained on the same coach-based split. The two sets of estimates correlated highly (Spearman $\rho = 0.81$), confirming that relative coaching rankings are robust to modeling choice. The mean coaching WAR across all team-seasons was -0.01 games (median = 0.06), confirming the median replacement baseline approximately centers the distribution near zero. Feature importance analysis confirmed coaching-specific features accounted for 42% of total model importance, indicating that coaches exert substantial influence on outcomes when controlling for roster quality, financial resources, and competitive context. Appendix C (Table 8) shows the top 20 features by importance.

4.1 Football Context and Career Leaders

While single-season WAR ranged from +5.9 to −8.6 (Section 4.6), career averages provide more reliable assessments owing to larger sample sizes and correspondingly narrower confidence intervals. Among coaches with at least 3 seasons, the top performers averaged between +1.2 and +2.3 WAR per season; the worst ranged from −3.3 to −1.9 WAR. Tables 1 and 2 report the best and worst coaches by average WAR, with 95% confidence intervals.

Long-tenured coaches accumulate enough seasons to yield precise estimates: Don Shula's +2.0 WAR per season (95% CI: [+1.5, +2.5]) over 26 seasons and John Madden's +2.1 per season (95% CI: [+1.3, +2.8]) over 9 seasons both have intervals entirely above zero. Shorter careers naturally produce wider intervals. Kevin O'Connell's +2.3 per season (95% CI: [+1.3, +3.2]) is entirely positive despite only three seasons, while coaches such as Gene Stallings and Red Miller have intervals that span zero, suggesting their point estimates may be influenced by individual standout seasons. Total WAR intervals capture both

Tab. 1: Top 15 Coaches by Average WAR per Season (minimum 3 seasons). Confidence intervals are computed from the t-distribution applied to each coach's individual-season WAR values (Equation 5).

Rank	Coach	Seasons	Total WAR	Total 95% CI	Avg. WAR	Avg. 95% CI
1	Kevin O'Connell	3	+6.8	[+3.8, +9.7]	+2.3	[+1.3, +3.2]
2	Gene Stallings	4	+9.0	[−0.6, +18.7]	+2.3	[−0.2, +4.7]
3	John Ralston	5	+10.9	[+5.2, +16.5]	+2.2	[+1.0, +3.3]
4	John Madden	9	+18.7	[+12.0, +25.4]	+2.1	[+1.3, +2.8]
5	Don Shula	26	+52.1	[+40.1, +64.0]	+2.0	[+1.5, +2.5]
6	Mike McDaniel	3	+5.4	[−3.3, +14.2]	+1.8	[−1.1, +4.7]
7	Kevin Stefanski	5	+8.4	[−2.3, +19.2]	+1.7	[−0.5, +3.8]
8	Chuck Pagano	6	+9.7	[−5.1, +24.5]	+1.6	[−0.8, +4.1]
9	Red Miller	4	+6.4	[−6.4, +19.2]	+1.6	[−1.6, +4.8]
10	Kyle Shanahan	8	+12.1	[+2.5, +21.7]	+1.5	[+0.3, +2.7]
11	Mike Vrabel	6	+8.6	[+1.3, +15.9]	+1.4	[+0.2, +2.6]
12	Mike Martz	6	+8.6	[+0.3, +16.8]	+1.4	[+0.1, +2.8]
13	Nick Sirianni	4	+5.7	[+0.7, +10.8]	+1.4	[+0.2, +2.7]
14	Bill Parcells	19	+25.4	[+15.0, +35.9]	+1.3	[+0.8, +1.9]
15	Bill Belichick	29	+34.0	[+12.1, +56.0]	+1.2	[+0.4, +1.9]

Tab. 2: Bottom 15 Coaches by Average WAR per Season (minimum 3 seasons). Confidence intervals are computed as in Table 1.

Rank	Coach	Seasons	Total WAR	Total 95% CI	Avg. WAR	Avg. 95% CI
1	Pat Shurmur	4	−13.3	[−18.6, −7.9]	−3.3	[−4.7, −2.0]
2	Steve Spagnuolo	3	−8.9	[−19.0, +1.2]	−3.0	[−6.3, +0.4]
3	Matt Eberflus	3	−8.7	[−17.8, +0.4]	−2.9	[−5.9, +0.1]
4	Hue Jackson	4	−11.4	[−25.7, +2.9]	−2.9	[−6.4, +0.7]
5	Jack Patera	7	−18.5	[−36.7, −0.2]	−2.6	[−5.2, −0.0]
6	Paul Wiggin	3	−7.8	[−19.3, +3.8]	−2.6	[−6.5, +1.3]
7	Scott Linehan	3	−7.8	[−17.3, +1.7]	−2.6	[−5.8, +0.6]
8	Rod Marinelli	3	−7.2	[−17.4, +2.9]	−2.4	[−5.8, +1.0]
9	David Shula	5	−11.6	[−28.6, +5.4]	−2.3	[−5.7, +1.1]
10	J.D. Roberts	3	−6.9	[−12.9, −0.9]	−2.3	[−4.3, −0.3]
11	Jim Schwartz	5	−10.8	[−20.4, −1.3]	−2.2	[−4.1, −0.3]
12	Bruce Coslet	8	−16.7	[−25.4, −7.9]	−2.1	[−3.2, −1.0]
13	Mike Nolan	4	−7.4	[−15.4, +0.6]	−1.9	[−3.9, +0.1]
14	Rick Forzano	3	−5.8	[−33.3, +21.8]	−1.9	[−11.1, +7.3]
15	Abe Gibron	3	−5.8	[−16.2, +4.6]	−1.9	[−5.4, +1.5]

per-season impact and career length: Shula's +52.1 wins (95% CI: [+40.1, +64.0]) represent the most precisely estimated career in the dataset.

Approximately 3.7 wins per season separates coaches at the 95th percentile from those at the 5th percentile, consistent with the three-to-four victory range reported by Hadley et al. (2000).

4.2 Coach WAR in Aggregate

Figure 1 illustrates the relationship between coaching longevity and WAR performance by plotting coaches based on average WAR per season and career length (measured in seasons coached).

The upper-right and lower-left quadrants show a positive relationship between coaching quality and career length, consistent with organizations retaining successful coaches. However, the lower-right quadrant contains coaches with 5+ seasons despite near-zero or negative WAR, suggesting possible suboptimal retention decisions. Context-adjusted evaluation may help identify such cases earlier.

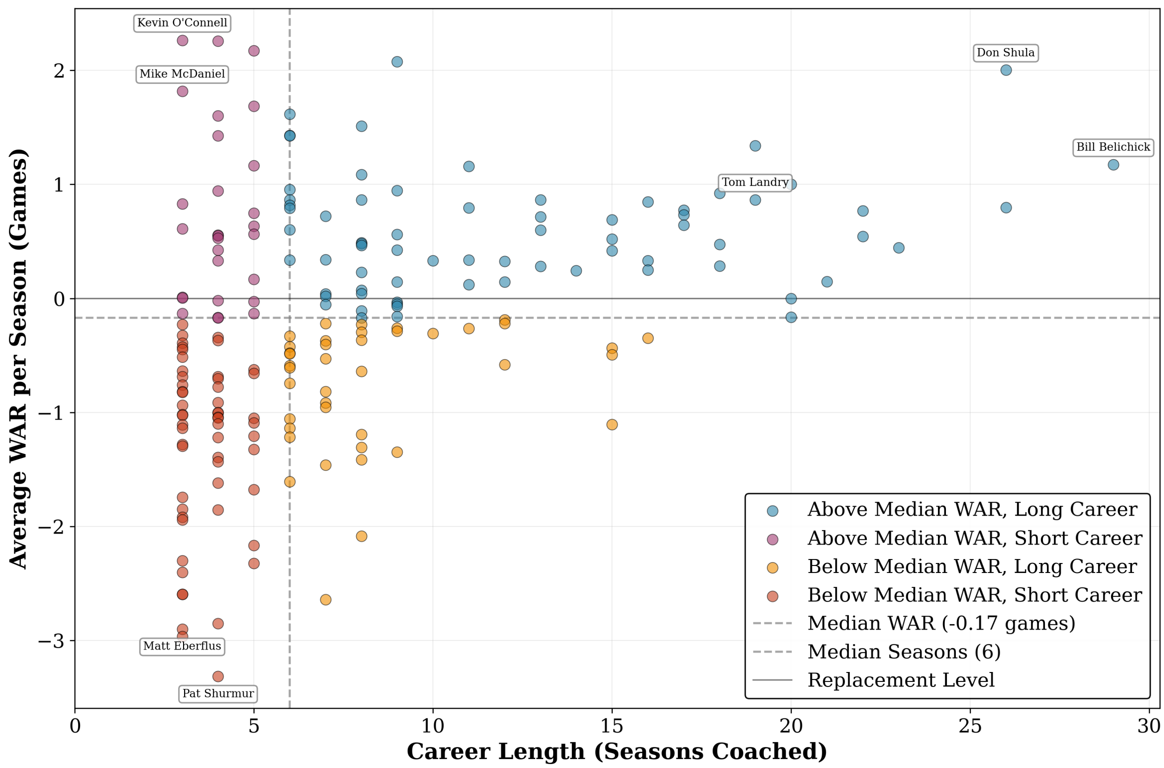

Fig. 1: Coach WAR Career Distributions. Coaches with two or fewer seasons are excluded.

4.3 Offensive versus Defensive Coaches

A recurring question concerns whether offensive or defensive coaching backgrounds produce more successful head coaches. For this analysis, coaches were classified based on their coordinator or position coaching experience prior to their head coaching appointment. Across the full modern era (1970–2024), no significant aggregate difference emerges between the two groups. However, this aggregate view obscures substantial era-level variation. Table 3 reports mean WAR by background and decade alongside Mann–Whitney U tests with 95% confidence intervals on the difference.

Tab. 3: Mann-Whitney U Test comparing Coaching Backgrounds by Decade. 95% CIs are Welch's t-intervals on the difference in means (Offensive − Defensive).

Metric	**1970's**	**1980's**	**1990's**	**2000's**	**2010's**	**2020's**
Coaches (O/D)	34/21	30/22	33/28	40/37	42/33	33/20
Seasons (O/D)	87/82	146/83	143/126	152/151	158/135	92/56
Mean WAR (O/D)	−0.571/+0.298	+0.311/+0.042	−0.482/+0.202	−0.148/−0.072	−0.006/+0.058	+0.260/−0.332
Difference (O − D)	−0.869	+0.269	−0.684	−0.077	−0.063	+0.592
95% CI	[−1.45, −0.29]	[−0.28, +0.82]	[−1.13, −0.24]	[−0.49, +0.34]	[−0.52, +0.40]	[−0.05, +1.24]
U-statistic	2676	6303	7300	11342	10547	2981
p-value	0.005	0.613	0.007	0.861	0.871	0.110
Significance	Highly Sig.	—	Highly Sig.	—	—	—

In the 1970s, defensive-background coaches held a significant advantage: the offensive-minus-defensive difference was −0.87 wins per season (95% CI: [−1.45, −0.29], $p = 0.005$). By the 2000s, the groups were indistinguishable (−0.08, 95% CI: [−0.49, +0.34], $p = 0.86$). In the 2020s, the direction has reversed, with offensive coaches averaging +0.59 WAR per season more than defensive coaches (95% CI: [−0.05, +1.24], $p = 0.11$). The 2020s interval includes zero, indicating that this reversal, while suggestive, has not yet reached conventional significance levels with the available sample. However, if the current performance differential persists through the remainder of the decade, the roughly doubled sample size would yield a projected $p \approx 0.02$, reaching conventional significance.

Cumulatively, offensive coaches have improved by +0.83 wins per season from the 1970s to the 2020s (95% CI: [+0.28, +1.38], $p = 0.004$), while defensive coaches have declined by −0.63 over the same period (95% CI: [−1.30, +0.04], $p = 0.064$).

4.4 WAR Persistence and Scheme Obsolescence

Coaching performance is unstable from year to year. Analyzing 1,163 consecutive season pairs from 1970–2023, Year N coaching WAR explains only 6.9% of the variance in Year N+1 performance (Figure 2). Top-quintile coaches averaging +2.58 WAR in Year N regress to +0.54 in Year N+1, retaining roughly 21% of their advantage. Bottom-quintile coaches at −2.79 WAR improve to −1.47 WAR after applying a 10th percentile penalty for mid-season terminations (Figure 3).

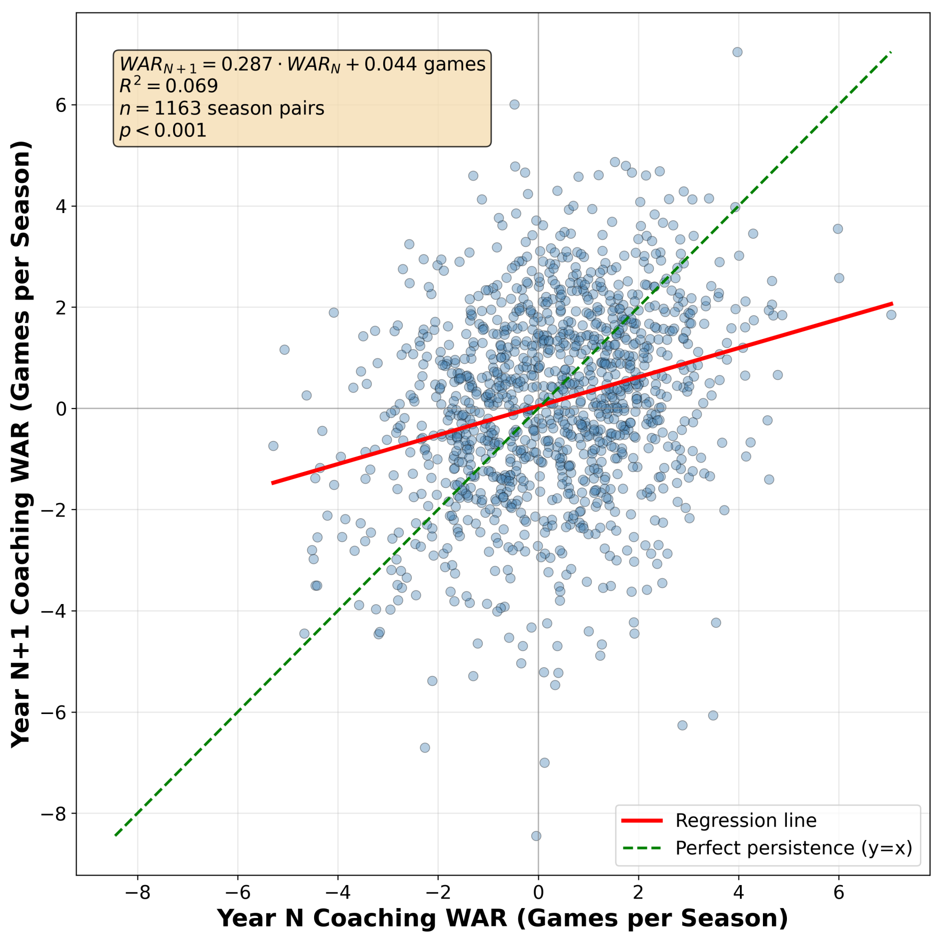

Fig. 2: Coach WAR in Year N+1 vs. Year N (persistence)

This volatility contrasts sharply with raw winning percentage, which shows approximately double the year-to-year stability ($R^2 = 0.137$, Appendix D, Figure 4). Winning percentage likely captures persistent

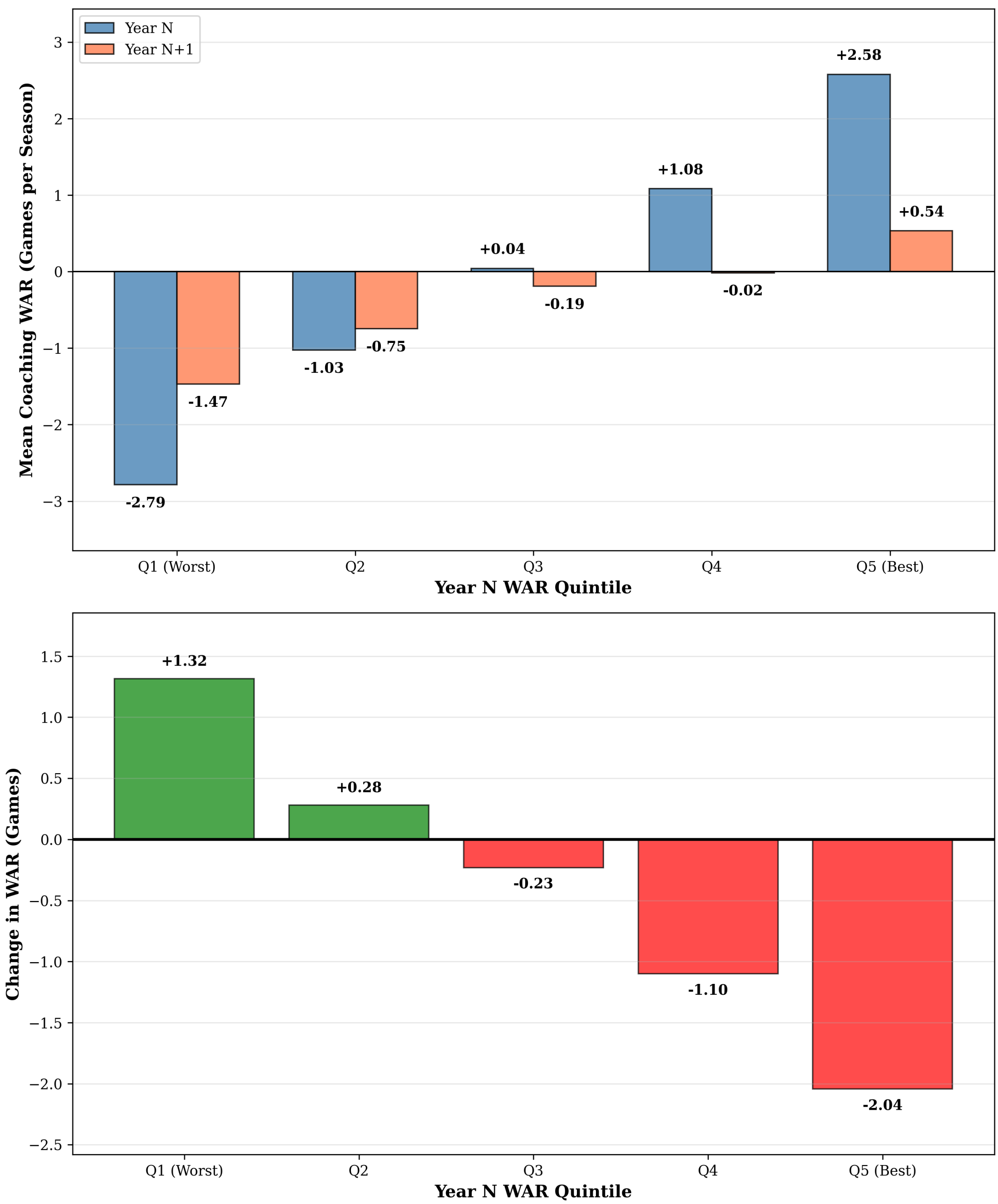

Fig. 3: Regression to the mean by WAR quintile. Top panel: mean WAR in Year N (blue) and Year N+1 (orange) for each quintile. Bottom panel: mean change in WAR from Year N to Year N+1. Bottom-quintile Year N+1 values apply a 10th percentile penalty for coaches terminated mid-season.

organizational advantages, such as market size, ownership stability, and accumulated roster talent, that carry forward regardless of coaching. The coach-specific contribution, by contrast, fluctuates considerably from season to season.

One interpretation of this low persistence is scheme obsolescence: coaching innovations that produce short-term advantages may be studied, copied, or countered by opposing staffs, reducing the advantage over time. Shared game film and a competitive market for coordinators and assistants provide plausible mechanisms for such diffusion. However, because WAR already conditions on roster composition and injury context, the residual volatility is specifically in the coaching component, though the inherent noise of a 16–17 game season limits the precision of single-year estimates.

The instability is not uniform across coaching backgrounds. While both offensive and defensive coaches show similar single-year persistence ($R^2 = 0.064$ and 0.092 respectively), multi-year averaging reveals a divergence. At a 2-year horizon, defensive coaches' predictive power approximately doubles to $R^2 = 0.180$, while offensive coaches remain flat at $R^2 = 0.069$. This difference is statistically significant (Fisher's Z $p = 0.009$; Appendix E, Figure 5 and Table 9). At three-year horizons, the gap persists ($p = 0.020$). These patterns suggest that offensive schemes may depreciate more rapidly than defensive approaches, consistent with the notion that offensive innovations are more easily studied and countered from game film.

4.5 Limitations and the "Franchise Quarterback Effect"

Tables 4 and 5 show the top and bottom 10 single-season WAR instances in the modern era.

Tab. 4: Top 10 Single-Season WAR Instances. Model residual standard error: ±2.1 wins per season.

Rank	Team	Year	Coach	Actual Win %	Replacement Win %	WAR	±1 SE
1	CHI	1987	Mike Ditka	0.733	0.364	+5.9	[+3.8, +8.0]
2	DET	2024	Dan Campbell	0.882	0.598	+4.5	[+2.4, +6.6]
3	SFO	2019	Kyle Shanahan	0.812	0.531	+4.5	[+2.4, +6.6]
4	CRD	1989	Gene Stallings	0.455	0.176	+4.4	[+2.3, +6.5]
5	PIT	2004	Bill Cowher	0.938	0.664	+4.4	[+2.3, +6.5]
6	NWE	2017	Bill Belichick	0.812	0.540	+4.4	[+2.3, +6.5]
7	MIN	2017	Mike Zimmer	0.812	0.540	+4.4	[+2.3, +6.5]
8	CLT	2012	Chuck Pagano	0.688	0.417	+4.3	[+2.2, +6.4]
9	GNB	1998	Mike Holmgren	0.688	0.424	+4.2	[+2.1, +6.3]
10	CLT	2009	Jim Caldwell	0.875	0.617	+4.1	[+2.0, +6.2]

Tab. 5: Bottom 10 Single-Season WAR Instances. Model residual standard error: ±2.1 wins per season. Asterisk (*) denotes mid-season termination.

Rank	Team	Year	Coach	Actual Win %	Replacement Win %	WAR	±1 SE
1	ATL	2020	Dan Quinn*	0.000	0.539	−8.6	[−10.7, −6.5]
2	SEA	1982	Jack Patera*	0.000	0.494	−7.9	[−10.0, −5.8]
3	PHI	1971	Jerry Williams*	0.000	0.493	−7.9	[−10.0, −5.8]
4	CIN	1996	David Shula*	0.143	0.581	−7.0	[−9.1, −4.9]
5	HTX	2020	Bill O'Brien*	0.000	0.426	−6.8	[−8.9, −4.7]
6	DAL	2010	Wade Phillips*	0.125	0.512	−6.2	[−8.3, −4.1]
7	DET	1976	Rick Forzano*	0.250	0.635	−6.2	[−8.3, −4.1]
8	CLT	1972	Don McCafferty*	0.200	0.537	−5.4	[−7.5, −3.3]
9	CIN	1971	Paul Brown	0.286	0.610	−5.2	[−7.3, −3.1]
10	CLT	1991	Ron Meyer*	0.000	0.313	−5.0	[−7.1, −2.9]

The highest single-season WAR values reveal an important limitation. Five of the top 10 single-season values coincide with elite or exceptional quarterback play (Tom Brady with Belichick in 2017, Peyton Manning with Caldwell in 2009, Brett Favre with Holmgren in 1998, rookie Ben Roethlisberger's 15–1 season with Cowher in 2004, and rookie Andrew Luck's turnaround with Pagano in 2012). This "franchise quarterback effect" reflects the inherent challenge of isolating coaching contributions: elite quarterbacks and elite coaching are rarely independent. While the framework controls for roster quality through player age, experience, and positional spending, it cannot fully disentangle whether a coach maximizes quarterback talent or whether an elite quarterback elevates apparent coaching performance. The remaining top-10 entries—Ditka with Jim McMahon starting only 6 of 15 games (1987), Campbell with Jared Goff (2024), Shanahan with Jimmy Garoppolo (2019), Stallings with Gary Hogeboom (1989), and Zimmer with backup Case Keenum (2017)—occurred without consensus elite quarterbacks, indicating the framework captures coaching performance beyond quarterback quality. Career-average analysis with confidence intervals (Tables 1–2) naturally mitigates single-season noise.

Nine of the ten bottom single-season WAR values (Table 5) correspond to mid-season terminations, with win percentages based only on games coached before dismissal. Scaling partial-season records to a full 16-game season may amplify negative WAR estimates, though the firing itself reflects an organizational judgment that performance was unsustainable.

Bill Walsh's career illustrates both genuine coaching development and single-season variability: his first season (1979, 2–14) produced −4.8 WAR, yet his best season (1987) reached +3.5 WAR—an 8.4-win swing that preceded a third Super Bowl victory. Walsh's early struggles giving way to a dynasty reflect that coaching ability is not static. The model residual standard error of ±2.1 wins per season (Section 3.5) indicates that individual-season WAR values carry substantial uncertainty. Career averages with confidence intervals, as presented in Tables 1–2, provide more reliable assessments and should be preferred for evaluation purposes.

Additional limitations warrant consideration. First, while the model controls for roster quality through age, experience, turnover, and salary allocation, these proxy measures cannot fully capture player talent; teams with similar demographic profiles may differ substantially in talent level. Second, organizational factors beyond our data, such as ownership stability, drafting competence, and facility quality, likely influence outcomes but remain unobserved.

4.6 Practical Applications

Context-adjusted coaching evaluation may inform hiring, retention, and contract decisions by separating coaching performance from inherited roster quality. For example, a coach with a .500 record but consistently positive WAR may be undervalued relative to a coach with a .600 record who inherited superior talent. Similarly, retention decisions benefit from objective performance baselines: cumulative negative WAR over multiple seasons, particularly when the confidence interval excludes zero (Table 2), provides quantitative evidence that performance deficits are structural rather than circumstantial.

5 Conclusion

This paper introduces a counterfactual framework for evaluating NFL head coach performance by comparing actual team outcomes to predictions under replacement-level coaching. Three principal findings emerge. First, coaching impact is substantial: approximately 3.7 wins per season separate coaches at the 95th percentile from those at the 5th percentile, with career leaders ranging from +2.3 WAR per season (95% CI: [+1.3, +3.2]) to −3.3 (95% CI: [−4.7, −2.0]). Second, the relative effectiveness of offensive and defensive coaching backgrounds has shifted markedly over the modern era, from a significant defensive advantage in the 1970s (−0.87, 95% CI: [−1.45, −0.29]) to a suggestive offensive advantage in the 2020s (+0.59, 95% CI: [−0.05, +1.24]). Third, coaching WAR exhibits low year-to-year persistence ($R^2 = 0.069$), with top-quintile coaches losing approximately 79% of their Year N advantage by Year N+1, consistent with scheme obsolescence as opponents adapt. Offensive coaches show significantly less multi-year persistence than defensive coaches (Fisher's Z $p = 0.009$ at the 2-year horizon).

Among several directions for future work, two appear particularly promising. First, replacing draft pick counts with expected value estimates based on draft position would better capture roster talent, as a first-overall selection carries substantially more value than a late-round pick within the same round. Second, a game-level WAR methodology, analogous to the play-level approach of Yurko et al. (2019), would enable bootstrap resampling for model-based confidence intervals on single-season point estimates rather than relying on residual-based approximations.

Data Availability

Coaching history data was scraped from Pro Football Reference. Processed data and analysis code are available at https://github.com/jwilliamson7/Coach_WAR or upon request to the corresponding author. Raw salary cap data from Spotrac is proprietary but processing scripts are provided.

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A Feature Count Across Categories

Tab. 6: Feature Count Across Categories

Feature Category	Coach Feature?	Number of Features
Metadata (excluded from training)	No	2
Salary Cap	No	19
Roster Turnover	No	63
Starter Turnover	No	27
Injury/Availability	No	37
Player Demographics	No	44
Penalty/Opponent	No	7
Draft Capital	No	29
Coaching Experience	Yes	7
Prior Team Performance (OC)	Yes	33
Prior Team Performance (DC)	Yes	33
Prior Team Performance (HC)	Yes	66
Target Variable	No	1
Total Columns in Dataset		**368**
Features Used in Training		**365**

B Selected Hyperparameters for the Final Model

Tab. 7: Selected Hyperparameters for the Final Model

Hyperparameter	Value
Objective	reg:squarederror
Random State	42
Number of Estimators	300
Learning Rate	0.05
Max Estimator Depth	2
Gamma	0
Lambda	0.1
Alpha	0.1
Subsample	0.7
Colsample by Tree	1.0
Minimum Child Weight	5

C Feature Importance Ranking of Top 20 Features

Tab. 8: Feature Importance Ranking of Top 20 Features

Rank	Feature Description	Importance	Coach Metric
1	Reserve/injured/suspended cap as percentage	0.049	No
2	Active 53-man roster cap as percentage	0.034	No
3	Number of new QBs added to roster	0.022	No
4	Normalized opponent avg pts per drive under HC	0.020	Yes
5	Normalized opponent interceptions thrown	0.019	No
6	Normalized team interceptions thrown	0.018	No
7	Normalized points scored under HC	0.016	Yes
8	Normalized opponent rushing first downs under HC	0.014	Yes
9	Average experience of all starters	0.014	No
10	Normalized opponent points scored under HC	0.014	Yes
11	Average percentage of games missed by QBs	0.013	No
12	Normalized scoring percentage under HC	0.012	Yes
13	Normalized opponent rushing yards under HC	0.011	Yes
14	Normalized opponent net yards per pass attempt under HC	0.011	Yes
15	Normalized avg points per drive under HC	0.010	Yes
16	Normalized opponent first downs under HC	0.010	Yes
17	Normalized passing TDs allowed under DC	0.010	Yes
18	QB new player rate percentage	0.010	No
19	Normalized opponent total yards under HC	0.009	Yes
20	Normalized opponent yards per play under HC	0.008	Yes

D Win Percentage Persistence

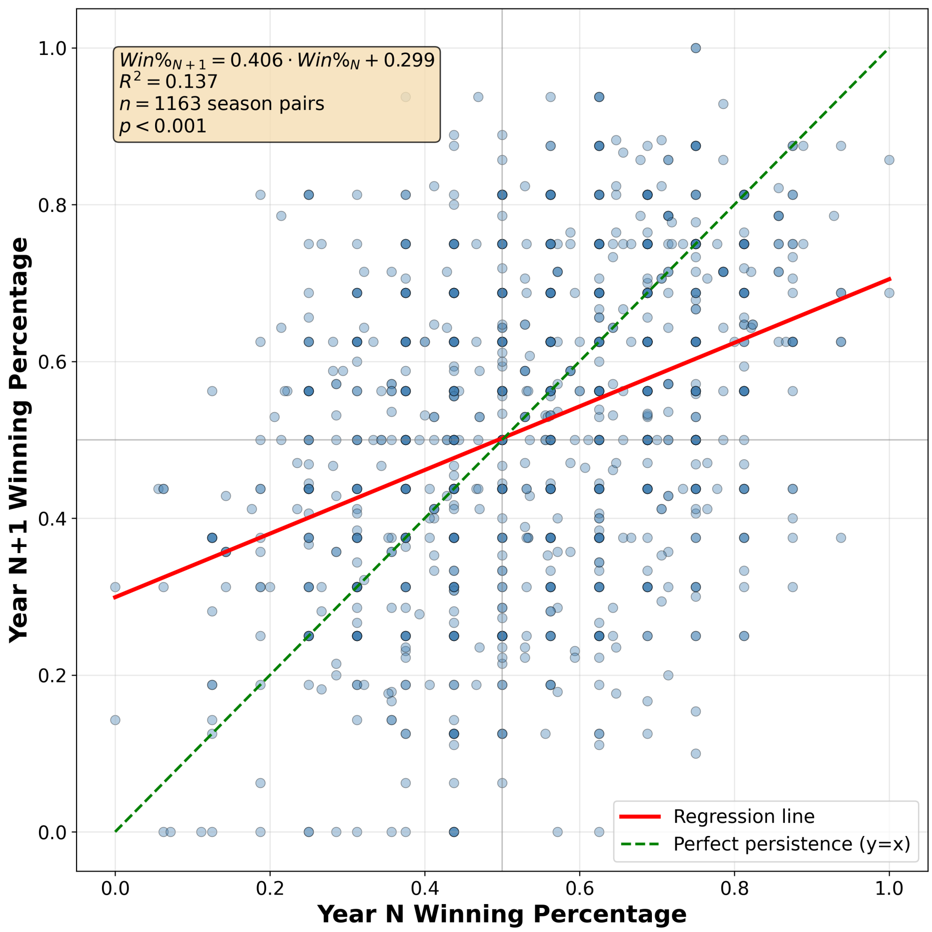

Fig. 4: Coach Win Percentage in Year N+1 vs. Year N

E WAR Persistence by Coach Background

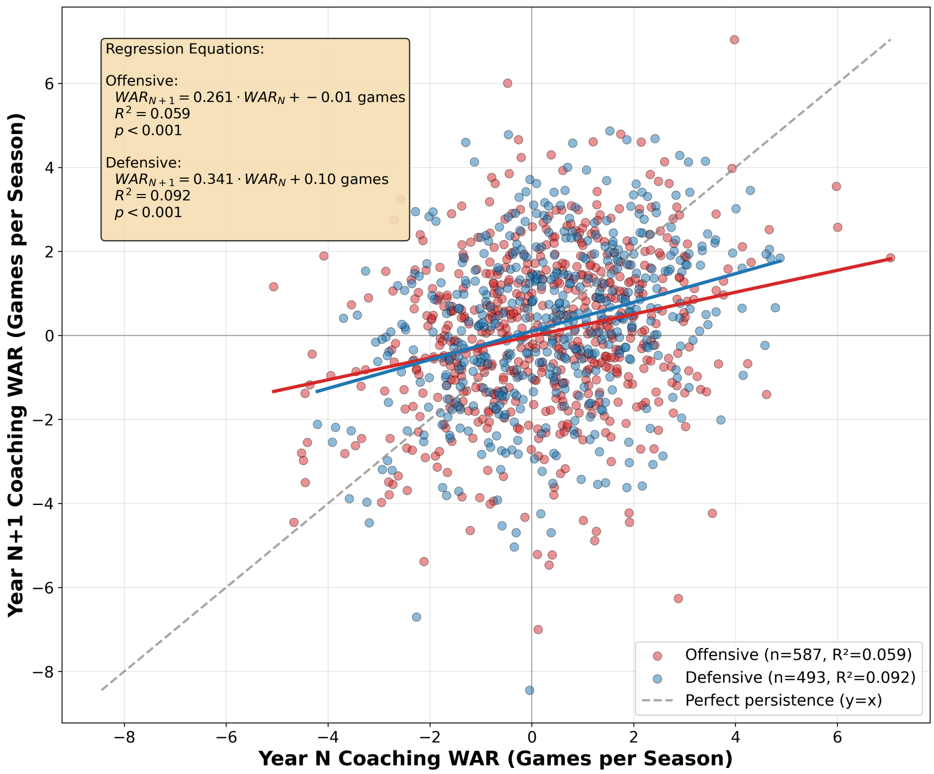

Fig. 5: Coach WAR in Year N+1 vs. Year N by Coach Background

Tab. 9: Coaching WAR Persistence by Background and Time Horizon

	Offensive		Defensive		Other	
Time Horizon	**R^2**	**p-value**	**R^2**	**p-value**	**R^2**	**p-value**
1-year	0.064	<0.001	0.092	<0.001	0.032	0.103
2-year avg	0.069	<0.001	0.180	<0.001	0.005	0.587
3-year avg	0.047	<0.001	0.153	<0.001	0.018	0.357
4-year avg	0.038	0.004	0.135	<0.001	0.019	0.429

Note: Fisher's Z-test for difference in correlations (Offensive vs Defensive): 1-year (p=0.380), 2-year (p=0.009), 3-year (p=0.020), 4-year (p=0.054).